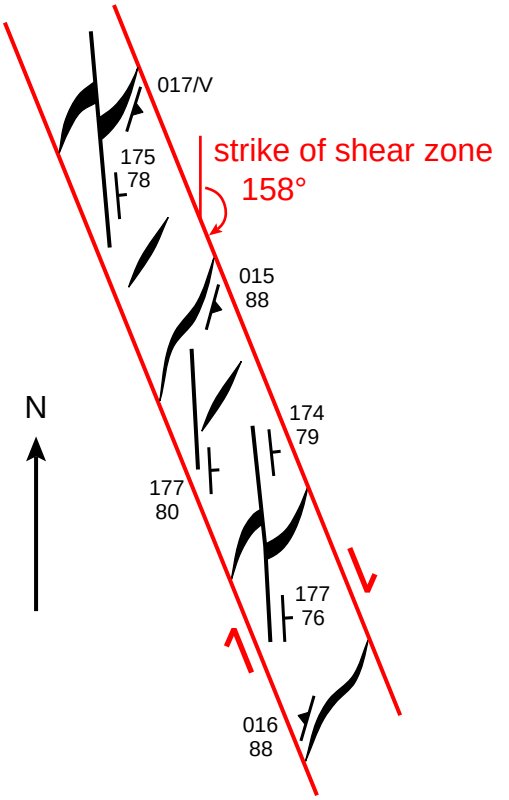
SHEAR ZONE EXERCISE

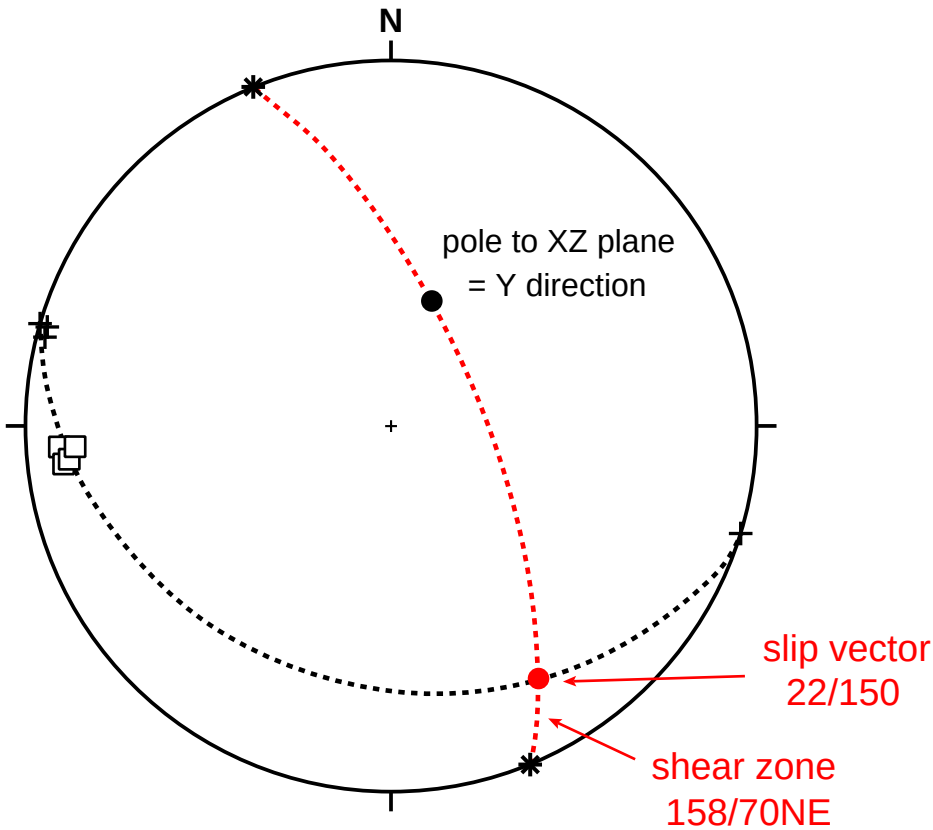
The map below shows the outcrop of a semi-brittle shear zone on a horizontal surface, together with measurements of en echelon extension veins and Riedel shears



Determine the orientation of the shear zone, the slip vector, and the sense of shear

1. Plot poles to extension veins and poles to Riedel shears and plot the great circle containing these. The pole to that great circle is the Y-direction which lies in the shear plane at 90° to the slip direction
2. On the map connect the tips of the sigmoidal extension veins and measure the trend of the shear zone; the plane containing this trend and the Y-direction is the shear plane
3. Count around 90° on the shear plane from the Y-direction and this is the slip vector
4. Determine sense of shear from orientation of slip vector and shear plane, noting that the sense of strike-slip follows from the obliquity of the Riedel shears (or extension veins) on the map

Shear sense is dextral and normal



✱ strike of shear zone
\+ poles to extension veins
□ poles to Riedel shears